MONASH University

MONASH BUSINESS SCHOOL

ETF3231/5231: Business forecasting

Week 12: Revision https://bf.numbat.space/

AACSB ACCREDITED
EQUIS ACCREDITED
ASSOCIATION OF MBAS AMBA ACCREDITED

Outline

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Assignment 1: forecast the following series

1. Google closing stock price on 20 March 2026.
2. Maximum temperature at Melbourne airport on 11 April 2026.
3. The difference in points (Collingwood minus Essendon) scored in the AFL match between Collingwood and Essendon for the Anzac Day clash. 25 April 2026.
4. The seasonally adjusted estimate of total employment for April 2026 in ('000). ABS CAT 6202, to be released around mid May 2026.
5. Google closing stock price on 22 May 2026.

For each of these, give a point forecast and an 80% prediction interval.

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Prize: $AU100 Amazon gift voucher

Forecasting competition: scoring

y = actual, $\hat{y}$ = point forecast, $[\hat{\ell}, \hat{u}]$ = prediction interval

Point forecasts:

$$\text{Absolute Error} = |y - \hat{y}|$$

- Rank results for all students in class
- Add ranks across all five items

Prediction intervals:

$$\text{Interval Score} = (\hat{u} - \hat{\ell}) + 10(\hat{\ell} - y)_+ + 10(y - \hat{u})_+$$

- Rank results for all students
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and the winners are…

and the winners are…

In third place: Jiayi Shao

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In second place: Dom

and the winners are…

In third place: Jiayi Shao

In second place: Dom

In first place:: Zonglin Cao

Assignment 1

Stock price forecasting (Q1 and Q5)

- Hard to beat naïve forecast
- Random walk model says forecast variance = $h\sigma^2$.

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Maximum temperature at Melbourne airport (Q2)

- Weather is relatively stationary over similar time of year and recent years.
- So take mean and var of max temp in April over last 10 years.

Difference in points in AFL match (Q3)

- Teams vary in strength from year to year.
- Could look at distribution of for-against points from last few years across all games for each team. Assume distributions independent.

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Difference in points in AFL match (Q3)

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Seasonally adjusted estimate of total employment (Q4)

- Probably locally trended.
- Perhaps use drift method based on average monthly change in last 2 years.
- Perhaps an ETS(A,A,N).

Outline

Topics

- Introduction to forecasting and R (1, Appendix)
- Time series graphics (2)
- Time series decomposition (3, 4)
- The forecasters' toolbox (5)
- Exponential smoothing (8)
- ARIMA models (9)
- Multiple regression (7)
- Dynamic regression models (10)

1. Introduction to forecasting and R

- Time series data and `tsibble` objects.
- What makes things hard/easy to forecast.

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Exam preparation

- Reading R code.
- Interpreting R output.

2. Time series graphics

- Time plots
- Seasonal plots
- Seasonal subseries plots
- Lag plots
- ACF
- White noise

3: Time series decomposition

- Describing a time series: seasonality, trend, cycles, changing variance, unusual features
- Transformations (and adjustments)
- Difference between seasonality and cyclicity
- Moving averages
- Classical and STL (advantages/disadvantages)
- Interpreting a decomposition
- Seasonal adjustment
- Forecasting and decomposition

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- Residual diagnostics: white noise, ACF, LB test.
- Problem of over-fitting.
- Out-of-sample accuracy. Training/test sets.
- Measures of forecast accuracy: MAE, MSE, RMSE, MAPE, MASE, RMSSE.
- Time series cross-validation.

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- Measures of forecast accuracy: MAE, MSE, RMSE, MAPE, MASE, RMSSE.
- Time series cross-validation.
- One-step prediction intervals based on RMSE from residuals.

8: Exponential smoothing

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- Holt’s local trend method.
- Damped trend methods.
- Holt-Winters seasonal method (additive and multiplicative versions).
- ETS state space formulation.

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- ETS state space formulation.
- Interpretation of output in R.
- Computing point forecasts by setting expectation of future ε_t to 0.
- Understanding how to generate prediction intervals.
- You have access to formula in the exam.

- Stationarity.
  - Transformations
  - Differencing (first- and seasonal-differences). What to use when.

9: ARIMA models

- Stationarity.
  - Transformations
  - Differencing (first- and seasonal-differences). What to use when.
- White noise, random walk, random walk with drift, AR(p), MA(q), ARMA(p,q), ARIMA(p, d, q), ARIMA(p, d, q)(P, D, Q)$_m$.
- ACF, PACF. Model identification.
- ARIMA models, Seasonal ARIMA models

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- ACF, PACF. Model identification.
- ARIMA models, Seasonal ARIMA models
- Order selection and goodness of fit (AICc)
- Interpretation of output in R.

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- Backshift operator notation.
- Expanding out an ARIMA model for forecasting.
- Finding point forecasts for given ARIMA process.

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- Backshift operator notation.
- Expanding out an ARIMA model for forecasting.
- Finding point forecasts for given ARIMA process.
- Understanding prediction intervals.
- One-step prediction intervals based on RMSE.
- Effect of (p,q) and (c,d) on forecasts.
- ARIMA vs ETS.

7: Multiple regression

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- Scenario forecasting.

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- Scenario forecasting.
- (Matrix formulation.)

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- Forecasting for dynamic regression models with ARIMA errors.

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Five Sections, all to be attempted.

A Respond to any four (out of six) topics, show your understanding with at least five key insights/explanations.

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- 5 working sheets
- Calculator.

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- Identify your weak points and practice them.
- Write your own summary of the material.
- Practice explaining the material to a class-mate.

Help available

- See us during the consultation times (for details refer to the website).
- Discuss on the Ed Discussion forum. I/we will monitor but will not answer every post. This is mostly for you to use between yourselves.

Useful resources for forecasters

Organization:

- International Institute of Forecasters.

Annual Conference:

- International Symposium on Forecasting
  - Montreal, Canada, June 28- July 1, 2026.
  - Student members ($45).

Journals:

- International Journal of Forecasting
- Foresight (the practitioner’s journal)

Links to all of the above at **www.forecasters.org**

IIF Best Student Award

- https://forecasters.org/programs/research-awards/students/
- US$100
- A certificate of achievement from the IIF
- One year free membership of the Institute with all attendant benefits. Subscriptions to:
  - the International Journal of Forecasting
  - the practitioner journal: Foresight
  - The Oracle newsletter

Discounts on conference and workshop fees, and links to a worldwide community of forecasters in many disciplines.

Happy forecasting

> Good forecasters are not smarter than everyone else, they merely have their ignorance better organised. *Anonymous*

Outline

1. Assignment 1
2. Review of topics covered
3. Exam
4. Student Evaluation of Teaching and Units

This week's quiz - please fill in your SETUs

SHARE CONFIDENTIAL FEEDBACK

Start your SETU surveys

SCAN TO COMPLETE
YOUR SETU SURVEYS

… or see the moodle page.

Could you please leave brief comments on each of the following:

1. Quizzes (did you find them useful, what would you change?)
2. Chatbot (did you find it useful, what would you change?)
3. Staggered assignments (did you find them useful, what would you change?)
4. Webpage (did you it useful, what would you change?)